

PhantomGrid -- Demo Day Quick Reference

Team ZeroIntent | CBI Hackathon 2026 | Keep off-screen during Q&A

KEY NUMBERS (know these cold)		HARDEST JUDGE QUESTIONS
Detection rate	95.3% (143/150 attackers caught)	Q: Why IsolationForest, not neural network? NNs need 100+ training samples. We enroll with 5. IF works at n=5 with no distributional assumptions -- it's the only viable choice at this scale. Q: Why not just use MFA? MFA authenticates once at login. PhantomGrid authenticates continuously. An attacker with stolen credentials + OTP still can't fake the victim's typing rhythm and navigation habits. Q: Your benchmark is synthetic -- is it real? Synthetic validates engine correctness. We also ran Layer 3 on the CMU real keystroke dataset (51 subjects, 400 reps each): AUC 0.639 on 5 enrollment samples -- honest floor, documented. Path to production: 15-20 enrollment sessions. Q: High FPR on CMU -- doesn't that mean it fails? CMU uses a 10-char password with intervals up to 1600ms. Our system is tuned for 5-digit PINs (85-120ms). Dataset mismatch + 5 samples = expected result. Full composite (L1+L2+L3) reduces FPR significantly vs L3 alone. Q: Mobile? Desktop POC only. Mobile needs a native SDK for touch event capture. The ML backend is signal-agnostic== it takes numerical vectors regardless of source. Mobile is next milestone. Q: What about the 7 false negatives? Sophisticated mimics who partially replicated rhythm (kr=0.08-0.15). Residual risk. Production mitigation: device binding -- even a perfect rhythm mimic must also match the enrolled device fingerprint. Q: GDPR / DPDP compliance? We store derived vectors only, no PII, no keystrokes, no account numbers. Satisfies DPDP Act 2023 data minimisation. RBI data localisation met when deployed India-region. Q: How does it scale? Inference <5ms. Stateless scoring layer scales horizontally. SQLite -> PostgreSQL is one SQLAlchemy line. Redis for baseline lookup in production. No shared mutable state between requests.
False positive rate	0.0% (0/150 legit users flagged)	
ROC AUC (synthetic)	1.000 (perfect separation)	
ROC AUC (CMU real)	0.639 (L3 only, 5 enrollment samples)	
Precision	100.0% (every flag = real attacker)	
F1 score	0.976	
False negatives	7 (sophisticated mimics, partial rhythm mimic)	
THRESHOLDS & WEIGHTS		
Fusion formula	L1 x 0.30 + L2 x 0.40 + L3 x 0.30	
ALLOW	composite < 60 (green, transaction proceeds)	
OTP step-up	composite 60-79 (amber, re-authenticate)	
BLOCK	composite >= 80 (red, transaction terminated)	
Contamination	0.10 (IsolationForest tolerance)	
Baseline window	20 sessions (sliding, ALLOW only)	
DTW -> risk	min(100, (distance / 180) x 100)	
Replay window	5 minutes, SHA-256 signature	
LAYER SIGNALS		
L1 CognitiveTrap (0.30) decoy_tap_count, amount_hesitations -> IsolationForest		
L2 IntentTrace (0.40) bene_dwell_ms, avg_amount_iki -> IsolationForest		
L3 RhythmLock (0.30) pin_vector (5 IKI ms) -> DTW vs enrolled baseline		
DEMO COMMANDS (have terminals open)		
Legitimate user	python demo_legit.py	
Attacker	python demo_attacker.py	
Replay attack	python demo_replay.py	
Backend (local)	uvicorn backend.main:app --reload --port 8000	
Railway URL	phantomgrid-production.up.railway.app/docs	
Demo user	demo_user (seeded synthetic baseline)	
Dashboard	dashboard/index.html?user_id=demo_user	

BEFORE YOU GO ON: Ping Railway 5 min before your slot | Open 3 tabs: NexaBank UI + Dashboard + /docs | Run demo_legit.py once to confirm backend is live